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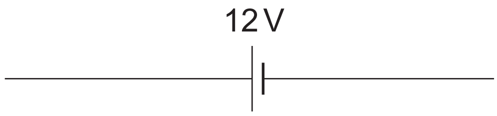
4. Jonathon is investigating how the resistance of a thermistor varies with temperature.

(a) Complete the circuit diagram used to measure the resistance of a thermistor. [3]



(b) Jonathon's results are shown in the table below.

Temperature (°C)	Current (mA)	Voltage (V)	Resistance (kΩ)
20	1.01	12.0	11.9
40	2.21	12.0	5.4
60	4.78	12.0	2.5
80	8.51	12.0	1.4
100	19.95	12.0	0.6

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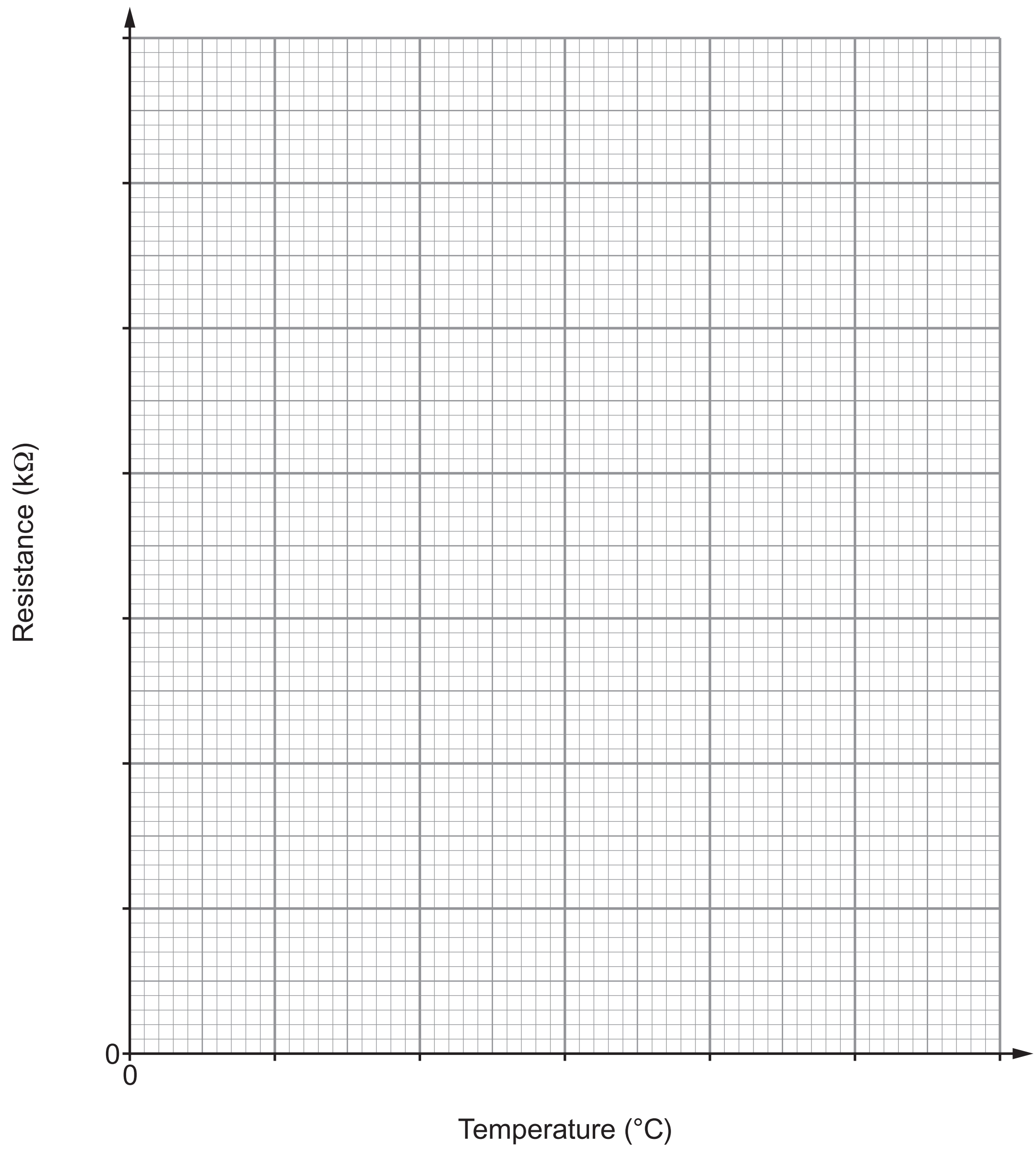
(i) Plot the data on the grid and draw a suitable line. [3]



(ii) Describe the relationship shown by the graph. [2]

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(iii) Use the graph and a suitable equation from the list below to answer each of the following questions.

$$\text{resistance } (\Omega) = \frac{\text{voltage (V)}}{\text{current (A)}}$$

$$\text{power} = \text{voltage} \times \text{current}$$

$$\text{power} = \text{current}^2 \times \text{resistance}$$

I. Calculate the current through the thermistor at 30 °C. [4]

current = ...mA

II. Calculate the power developed in the thermistor at 30 °C. [2]

power = ...W

(c) Thermistors are found in refrigerators to control the temperature.

Describe how the experiment needs to be modified to test if this thermistor would be suitable for a refrigerator. [2]

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